

Grade 6
Unit 3
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. Felix drove to Canada and noticed that the speed limit was 90 kilometers per hour. His speedometer said that he was driving 52 miles per hour. If 1 mile is approximately 1.609 kilometers, how many kilometers under the speed limit was Felix driving?

$$52 \times 1.609 = 83.668$$

$$90 - 83.668 = 6.332 \text{ km under the speed limit}$$

2. Suppose Felix claimed that he was driving 38 kilometers under the speed limit because $90 - 52 = 38$. Explain to Felix why he is incorrect.

Because 52 is measured in miles.

3. Vallaire had $3\frac{1}{3}$ bags of candy to put out for Halloween. Each bag had $2\frac{3}{4}$ pounds of candy. In the first hour, Vallaire gave $\frac{1}{3}$ of the candy to trick-or-treaters. How much candy, in pounds, did Vallaire give to trick-or-treaters in the first hour?

$$3\frac{1}{3} \times 2\frac{3}{4} \rightarrow \frac{10}{3} \times \frac{11}{4} = \frac{110}{12} = \frac{55}{6}$$

$$\frac{55}{6} \times \frac{1}{3} = \frac{55}{18} \text{ or } 3\frac{1}{18} \text{ pounds}$$

4. Explain why your answer to question 3 makes sense in the context.

Answers may vary. $\frac{55}{6}$ is a little greater than 9, so a third of that should be just over 3.

5. On Saturday, Jorge did homework for $1\frac{3}{4}$ hours. On Sunday, Jorge did homework for $\frac{4}{5}$ of the time that he did on Saturday. How many total hours of homework did Jorge do over the weekend?

$$\text{Sunday: } 1\frac{3}{4} \times \frac{4}{5} \rightarrow \frac{7}{4} \times \frac{4}{5} = \frac{28}{20} = \frac{7}{5}$$

$$\text{Total: } \frac{7}{4} + \frac{7}{5} \rightarrow \frac{35}{20} + \frac{28}{20} = \frac{63}{20} = 3\frac{3}{20} \text{ hours}$$

6. Gillian went to the grocery store and bought 2.4 pounds of hamburger meat for \$6.95 per pound. She paid with a \$20 bill. How much change did she receive?

$$20 - 2.4 \times 6.95$$

$$20 - 16.68 = \$3.32$$

7. Suppose Gillian thought she should get \$10.65 in change because $2.4 + 6.95 = 9.35$ and $20 + 9.35 = 10.65$. Explain to Gillian why she is incorrect.

Answers may vary. 2.4 is not money, it is the weight of the meat.

8. For the first 15 days in January, there was a total of $8\frac{3}{8}$ inches of snow in Chicago. On the 16th, it snowed $\frac{2}{5}$ of an inch every hour for 4 hours. What was the total snowfall for the first 16 days in January?

$$8\frac{3}{8} + 4 \times \frac{2}{5}$$

$$\frac{67}{8} + \frac{8}{5} \rightarrow \frac{335}{40} + \frac{64}{40} = \frac{399}{40} = 9\frac{39}{40} \text{ inches}$$

9. Explain why your answer to question 8 makes sense.

Answers will vary.

10. Marcello bought a plant that was 5.24 centimeters tall. If it grows approximately 0.22 centimeters per day, how many days will it be before the plant is 7 centimeters tall?

$$7 - 5.24 = 1.76$$

$$1.76 \div .22 = 8 \text{ days}$$